

Zihan Fang

zihan.fang@vanderbilt.edu | (615) 853-1277 | Website | LinkedIn

Research Focus

My research develops human-centered AI to lower barriers to acquiring expertise and contributing effectively to software development. I combine large-scale repository mining, cognitive and behavioral modeling, and reinforcement learning to identify sources of difficulty, characterize how developers acquire expertise, and translate these insights into AI-supported programming systems and immersive learning environments that improve learning efficiency, programming performance, and long-term skill development.

Education

Vanderbilt University, Ph.D. in Computer Science	Aug 2023 – Present
Vanderbilt University, M.S. in Data Science; GPA: 3.9/4.0	Aug 2021 – May 2023
Soochow University, B.S. in Information Management; GPA: 3.8/4.0	Sep 2016 – Jun 2020

Publications

- [1] **Zihan Fang**, Janice Chung, Ruijia Chen, Marcia Barnes, Yuhang Zhao, Yu Huang: *Toward Inclusive Programming Support for Novice Programmers with Dyslexia: Insights from a Comparative Study*. In the *Proceedings of the 41st IEEE/ACM International Conference on Automated Software Engineering (ASE 2026)*.
- [2] Ningzhi Tang, Chaoran Chen, **Zihan Fang**, Gelei Xu, Maria Dhakal, Yiyu Shi, Collin McMillan, Yu Huang, Toby Jia-Jun Li: *Programming by Chat: A Large-Scale Behavioral Analysis of 11,579 Real-World AI-Assisted IDE Sessions*. In the *Proceedings of the 41st IEEE/ACM International Conference on Automated Software Engineering (ASE 2026)*.
- [3] **Zihan Fang**, Weizhe Jiao, Robert Tairas, Yu Huang: *Novices Comprehending Functional vs. Imperative Programs: An Eye-Tracking Study*. *ACM Transactions on Software Engineering and Methodology (TOSEM)*.
- [4] Yueke Zhang, **Zihan Fang**, Kevin Leach: *Stop the Retrieval Thrash: Brain-Guided Episodic Memory for Repository-Scale Coding Agents*. In *Companion Proceedings of the ACM on Software Engineering*, Vol. 3, *FSE (FSE 2026, HumanAISE)*.
- [5] Yueke Zhang, Yifan Zhang, **Zihan Fang**, Greg Trafton, Daniel Levin, Kevin Leach, Yu Huang: *ScanCoder: Leveraging Human Attention Patterns to Enhance LLMs for Code*. In *Proceedings of the ACM on Software Engineering*, Vol. 3, *FSE (FSE 2026)*.
- [6] Yueke Zhang, **Zihan Fang**, Kevin Leach, Yu Huang: *Locating Software Vulnerabilities With Static Analyzers: How Far Are We?* In *Proceedings of the 30th International Conference on Evaluation and Assessment in Software Engineering (EASE 2026)*.
- [7] **Zihan Fang**, Robert Wallace, Zachary Karas, Toby Jia-Jun Li, Collin McMillan, Yu Huang: *Investigating the Feasibility of Conducting Webcam-Based Eye-Tracking Studies in Code Comprehension*. *IEEE Transactions on Software Engineering (TSE)*.
- [8] **Zihan Fang**, Yueke Zhang, Thomas Zimmermann, Denae Ford, Yu Huang: *Contribution Patterns in Open Source Software for Social Good: Dynamics, Individuals, and Impact*. In *Proceedings of the ACM on Human-Computer Interaction (CSCW 2026)*.
- [9] Yueke Zhang, **Zihan Fang**, Greg Trafton, Daniel Levin, Kevin Leach, Yu Huang: *CodeACT-R: A Cognitive Simulation Framework for Human Attention in Code Reading*. In *Companion Proceedings of the 40th IEEE/ACM International Conference on Automated Software Engineering (ASE 2025, NIER Track)*.
- [10] **Zihan Fang**, Jiliang Li, Anda Liang, Gina R. Bai, Yu Huang: *A Comparative Study on ChatGPT and Checklist as Support Tools for Unit Testing Education*. In *Companion Proceedings of the 33rd ACM International Conference on the Foundations of Software Engineering (FSE 2025, SEET Track)*.
- [11] **Zihan Fang**, Yu Huang: *“Math Is a Pain!”: Understanding Challenges and Needs of the Machine Learning Community on Stack Overflow*. In *Proceedings of the ACM on Human-Computer Interaction*. (*CSCW 2024*)
- [12] **Zihan Fang**, Madeline Endres, Thomas Zimmermann, Denae Ford, Westley Weimer, Kevin Leach, Yu Huang: *A Four-Year Study of Student Contributions to OSS vs. OSS4SG with a Lightweight Intervention*. In *Proceedings of the 31st ACM Joint European Software Engineering Conference and Symposium on the Foundations of Software Engineering (ESEC/FSE 2023, ACM SIGSOFT Distinguished Paper Award.)*

Manuscripts Under Review

- [1] **Zihan Fang**, Yifan Zhang, Yueke Zhang, Kevin Leach, Yu Huang: *DPO-F+: Aligning Code Repair Feedback with Developers’ Preferences*. [arXiv]
- [2] **Zihan Fang**, Yueke Zhang, Yu Huang: *AgentForge: An Immersive Role-Playing Platform for Learning Agentic Software Engineering*. [arXiv]

[3] **Zihan Fang**, Yueke Zhang, Ningzhi Tang, Collin McMillan, Toby Jia-Jun Li, Yu Huang: *From Conversation to Contribution: Characterizing Coding Agent in Open-Source Software*. [arXiv]

[4] Yueke Zhang, **Zihan Fang**, Kevin Leach, Yu Huang: *GraphAlignCoder: Aligning Program and Proof Graphs for Code Generation*. [arXiv]

Research and Industry Experience

Research Assistant, Vanderbilt University	Aug 2023 – Present
Conduct research on human-centered AI for software engineering, spanning developer-feedback alignment, AI-assisted programming, and multi-agent learning environments.	
Data Scientist, Tencent Inc.	Aug 2021 – Dec 2021
Built scalable ML and data systems to support real-time identity verification and model development.	
Data Analyst Intern, DiDi Inc.	Jan 2021 – Jun 2021
Applied data analytics and machine learning to segment drivers, evaluate operational strategies, and optimize recruiting, onboarding, and business development efforts.	

Teaching Experience

Teaching Assistant, Vanderbilt University	Aug 2022 – Present
- CS3892/5892 Designing Computing for Learning – Fall 2026 - DS3100 Fundamentals of Data Science – Fall 2022, Spring 2023 - CS3265 Database Management Systems – Fall 2023 - CS4278/5278 Principles of Software Engineering – Spring 2024	

Awards

SIGSOFT CAPS ASE Travel Award	2026
Ignite Grant at Vanderbilt University	2025
LIVE Horizon Grant at Vanderbilt University	2025
C. F. Chen Best Paper Award in the Department of Computer Science at Vanderbilt University	2024
Richard Bennett/Dorothy Danforth Compton Prize Scholarship at Vanderbilt University	2024
SIGSOFT Distinguished Paper Award at FSE	2023
NSF Student Travel Award	2023

Service

AAAI 2026: PC Member

Empirical Software Engineering (EMSE): Reviewer

SIGCSE TS 2025: PC Member (Posters; Computing Education Research)